STINet: Vegetation Changes Reconstruction Through a Transformer-Based Spatiotemporal Fusion Approach in Remote Sensing

Sai Wang and Fenglei Fan

***Abstract*— Filling gaps in high-resolution satellite imagery is essential for tracking vegetation changes over time. Spatiotemporal fusion (STF) aims to create fusion products that improve both spatial resolution and temporal coverage by using images from various remote sensing sources. However, most existing STF methods rely on the assumption that reflectance values for the same land-cover type remain constant between base and prediction dates, a premise often invalidated by the variability in vegetation disturbance and recovery scenarios, where differences in disturbance intensity, patterns, and phenological stages challenge this uniformity. Therefore, we propose a novel Transformer-based method, the spatiotemporal integration network (STINet), which is effective in fusing multiscale spatiotemporal dynamic features. STINet is structured around three key components. The feature fusion (FF) block effectively integrates multiscale spatiotemporal information into a deep learning (DL) framework. The adaptive feature extraction (AFE) block significantly improves the precision of pixel-level features, essential for detecting subtle changes in diverse vegetation patterns. The spatiotemporal-wise multihead self-attention (ST-MSA) module through its innovative self-attention mechanism across spatiotemporal dimensions, facilitated the reconstruction of vegetation dynamics. To verify the effectiveness and robustness of the proposed method, we conducted experiments in three carefully selected scenarios using multisensor and multitemporal imagery to reconstruct the dynamic changes in vegetation due to various disturbances and recovery processes. Compared to the four typical fusion methods [enhanced spatial and temporal adaptive reflectance fusion model (ESTARFM), flexible spatiotemporal data fusion method (FSDAF), extended super-resolution convolutional neural network (ESRCNN), and multiscene spatiotemporal fusion network (MUSTFN)], STINet achieved the best performance in preserving both spatial texture and spectral value. Furthermore, we showcased the applicability and effectiveness of STINet in accurately capturing phenological changes and distinguishing various farming activities.**

***Index Terms*— Multisensor satellite data, spatiotemporal fusion (STF), time-series data, transformer.**

I. Introduction

SATELLITE-BASED Earth observation is a key tool for tracking and analyzing surface changes on Earth, especially in vegetation [1]. Time-series analysis in this field allows for the observation of ongoing surface dynamics, which is crucial for monitoring changes in forest cover [2], urban and agricultural expansion [3], and vegetation dynamics response to climate changes [4]. However, acquiring highly spatial resolution and temporal resolution time-series remote sensing images remains a challenge due to atmospheric conditions like cloud cover and aerosols and satellite revisit period [5]. Integrating data from different sensors into a consistent time series requires meticulous preprocessing and calibration to ensure that the observed changes accurately reflect actual surface alterations and are not skewed by sensor inaccuracies [6]. To overcome these challenges, multisensor time series fusion techniques have been developed, creating virtual constellations for collaborative observation, aiming for a temporal density comparable to coarser-resolution systems [7]. This advancement marks a significant milestone in the field of Earth observation, significantly enhancing our ability to monitor the dynamic landscape of our planet accurately and comprehensively.

The spatiotemporal fusion (STF) technique aims to merge data from at least two satellite sensors, producing imagery with both fine spatial and enhanced temporal resolutions. By integrating coarse resolution, high temporal frequency data (e.g., MODIS) with high resolution, and low temporal frequency data (e.g., Landsat, Sentinel-2, Gaofen-5), STF achieves synthesized outputs that excel in spatial detail and temporal continuity [8], [9]. These methods are broadly classified into three categories: including weight function-based, unmixing-based, and learning-based. The weighted function-based methods primarily use geostatistical spatial–spectral interpolation techniques. These methods create weighted functions by analyzing the autocorrelation in observational data, which are then used to estimate values at unobserved points. A prime example is the spatial and temporal adaptive reflectance fusion model (STARFM) [7]. STARFM enhances image resolution by combining data from sensors like Landsat and MODIS, assuming that low-resolution MODIS pixels can be generated from high-resolution Landsat pixels by disregarding geographical positioning errors and atmospheric variances. However, the relationship between MODIS and Landsat reflectance is influenced by several factors, such as the presence of mixed pixels in MODIS data, changes in land cover types, and variations in the bidirectional reflectance distribution function (BRDF). To address these issues, the spatial–temporal adaptive algorithm for mapping reflectance change (STAARCH) [10]

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extracts indices of brightness, greenness, and wetness from Landsat and MODIS images using Tasseled Cap transformations, thereby distinguishing spatial variations with Landsat data and temporal changes with MODIS data. Subsequently, the Enhanced STARFM (ESTARFM) [11] was introduced. This model incorporates conversion coefficients to adjust variations in reflectance and utilizes all spectral bands to identify and select pixels with similar characteristics, thereby further improving the accuracy of the fusion outcomes. The algorithms of weight function-based methods are simple, ignoring the nonlinear changes and abrupt changes of the land surface, failing to capture surface heterogeneity over large spatial extents and long durations. The unmixing-based methods employ linear spectral mixing theory to decompose mixed pixels and deduce the values of individual components. As an early approach in this category, the multisensor multiresolution technique (MMT) [12] was proposed. This technique aims to calculate endmember proportions for larger pixels and assign unmixing reflectance values to smaller ones. The flexible spatiotemporal data fusion method (FSDAF) [13] integrates spectral unmixing with thin plate spline interpolation techniques, effectively capturing changes in land cover. However, this method assumes constant reflectivity between image pairs, and its prediction accuracy decreases under prolonged cloud cover. To further enhance the process, the variation-based spatiotemporal data fusion (VSDF) [14] method was developed, integrating feature-level information, and effectively utilizing the reliability of coarse-resolution images for precise reconstruction of fine-resolution images. Additionally, AgriFuse [15], an innovative category-based linear regression model, was designed for accurate estimation of unmixing model parameters, thereby achieving greater refinement in the fusion process. The unmixing method for solving regression coefficients is performed globally and may not be suitable for large regions with high spatial heterogeneity.

Traditional multisensor time-series fusion methods are lauded for their computational efficiency. Yet, they encounter significant challenges in accurately reconstructing time-series scenarios involving vegetation disturbances and regrowth, particularly in contexts characterized by rapid shifts in land surface dynamics. These scenarios often present diverse characteristics, including linear or nonlinear evolutionary trends, varying amplitudes, phases that differ markedly, or discernible breakpoints [16], [17]. Many traditional data fusion methods predominantly rely on the simulation of historical image data as constraints within the fusion process [18]. The absence of spatial constraints compounds this challenge. Spatial constraints are crucial for ensuring spatial coherence, and their lack can cause significant inconsistencies in fusion outcomes, particularly in regions subject to rapid and multifaceted environmental alterations.

In the last decade, learning-based approaches such as dictionary-pair learning, and deep learning (DL) have seen significant advancements in the field of remote sensing. These methodologies have been instrumental in implicitly establishing nonlinear mappings between fusion results and input datasets [19]. A prime example is the extended super-resolution convolutional neural network (ESRCNN), which employs a DL framework to address spatial resolution discrepancies between Landsat-8 and Sentinel-2 imagery [20]. Recently, the image fusion method that does not rely on fine reference images has attracted widespread attention; with its spectral and spatial attention mechanism, it is capable of accurately predicting multiscale information [21], [22]. Further developments include the generative adversarial network for one prior image pair case (OPGAN) [23]. This network uses a generative adversarial model and a discriminator within a min-max optimization framework to enhance the prediction of fine-resolution images, thereby improving fusion quality. Additionally, the multiscene spatiotemporal fusion network (MUSTFN) [24], effectively leverages multiscale features from images acquired via multiple sensors at varying resolutions to augment fusion effectiveness. Innovations also encompass a hybrid supervised and self-supervised learning framework [25], designed to tackle the challenges arising from significant spatiotemporal variations between auxiliary and prediction dates. In addition, a two-stage multiscale fusion module was proposed, where one stage fuse feature and another stage fuses contextual information between different scales [26]. While DL-based methods have gained popularity for their high prediction accuracy and processing efficiency, they are not devoid of limitations. A significant challenge lies in their limited capability to detect nonlocal self-similarity and long-range dependencies. This frequently results in failure to recognize essential correlations between the spatiotemporal characteristics of high-resolution images and the spectral attributes of their low-resolution counterparts.

In recent developments, Transformer networks, initially conceptualized for natural language processing (NLP), have demonstrated substantial potential in the field of computer vision. They excel at capturing extensive information and delivering accurate pixel-level details [27], [28]. Particularly, the multihead self-attention (MSA) mechanism within these models has proven effective in identifying nonlocal relationships and comprehensive relational patterns. This capability presents significant prospects for exploring global dependencies in spatiotemporal data fusion, especially when integrated with convolutional neural network (CNN) based methodologies. The significant scale disparity among spatiotemporal dynamic information from diverse sensors presents a formidable challenge in accurately representing spatiotemporal dynamics. This discrepancy impedes the efficacy of STF which relies on integrating temporal change features from MODIS with spatial change features from Landsat. Additionally, the pronounced differences in spatial sparsity across channels suggest that the efficiency in capturing spectral interactions may not match the efficacy in modeling spatial variation correlations [29], [30]. Furthermore, traditional Transformer networks employ a linear mapping approach for transforming input tokens into queries, keys, and values, followed by matrix multiplications for MSA computations. While this approach processes all tokens simultaneously, it may inadvertently neglect the critical spatiotemporal correlations. The existence of these problems leads to unavoidable performance

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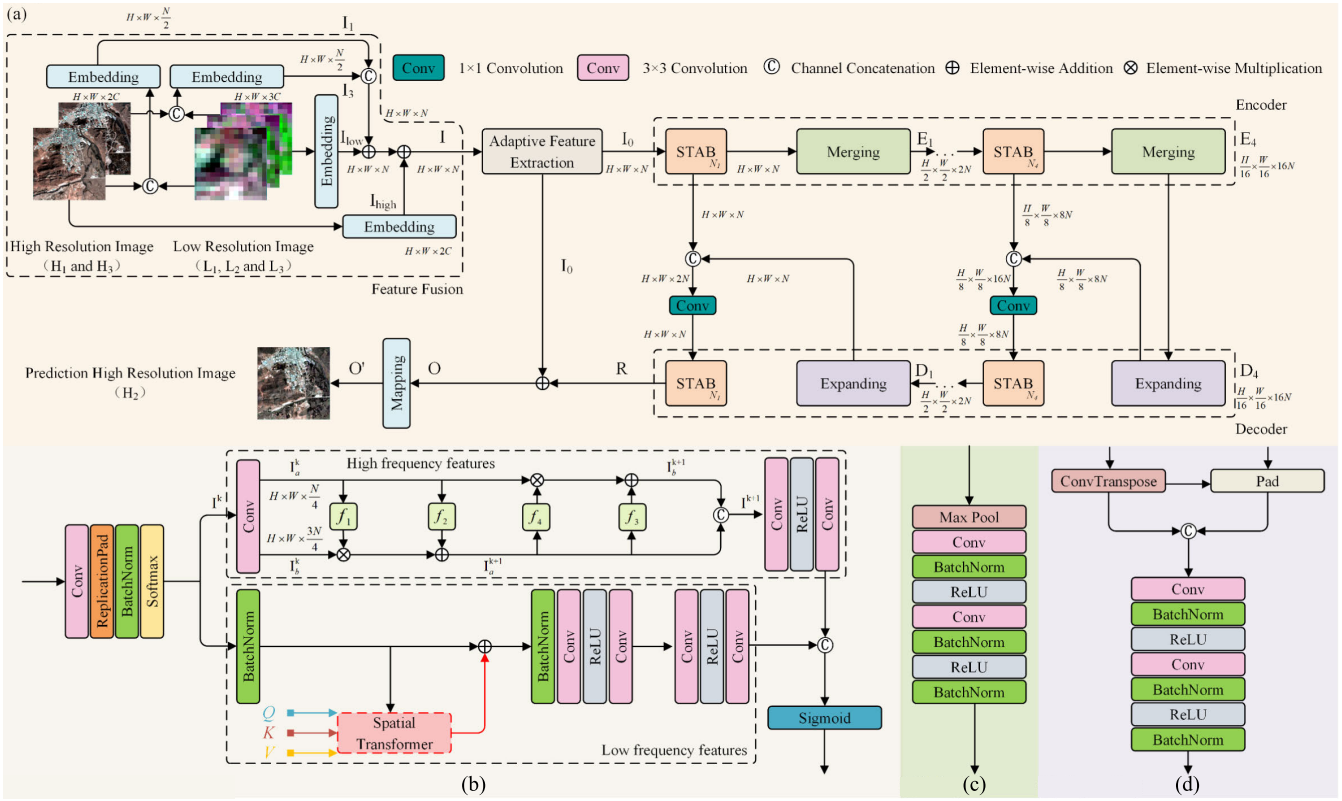


Fig. 1. Framework of our method. (a) Architecture of STINet and the FF mechanism. (b) Components of STINet fundamental unit, the FIE block. (c) and (d) Merging and expanding modules.

degradation, stemming from an insufficient capacity to learn long-range dependencies and a heightened risk of texture distortion in generating the fusion image.

To overcome the complexities in generating high-resolution imagery from multisensor data, we introduced the spatial–temporal integration network (STINet), a novel framework tailored for producing high spatiotemporal resolution surface reflectance images and normalized difference vegetation index (NDVI). STINet integrates three key components: the feature fusion (FF) mechanism, adaptive feature extraction (AFE) block, and spatiotemporal-wise MSA (ST-MSA) module, each playing a pivotal role in enhancing image fusion quality. The FF block uniquely links imagery across various timeframes, maintaining multiscale characteristics without downscaling, thereby offering an advantage over traditional approaches. In contrast, the AFE block, featuring adaptive feature filtering (AFF) and frequency information extraction (FIE), skillfully processes multilevel and scale features, adeptly responding to image and pixel variations. It enriches learning by harmonizing diverse feature sizes and adapting to image dynamics, with the FIE mechanism efficiently parsing essential shallow features through an advanced long-range attention strategy. The innovation culminates with the ST-MSA module, which processes each spatiotemporal feature as a distinct entity, employing self-attention along spatiotemporal dimensions. This sophisticated approach empowers STINet to precisely capture and reconstruct the intricate spatial patterns of sudden land cover change in varied terrains.

II. Methodology

In our endeavor to enhance the accuracy of terrestrial representations in satellite remote sensing imagery, we have developed STINet, utilizing a novel U-shaped architecture that synergistically consists of encoder, decoder, and skip connections. This architecture is anchored by the ST-MSA module, a key innovation in our approach. Our technical exposition, illustrated in Fig. 1(a), details the FF technique and the comprehensive network architecture of STINet. Further, Fig. 1(b) delves into the pivotal FIE module within STINet. To provide an in-depth view of our image processing methodology, Fig. 1(c) and (d) depict the critical Merging and Expanding modules, respectively. These modules are fundamental in the synthesis of satellite imagery that accurately mirrors ground truths, thereby forming an integral part of our image processing framework.

A. Overall Network Architecture

In the design of STINet, we leverage ST-MSA Blocks (STABs) as the foundational elements. The initial phase involves a FF mechanism, which concatenates multitemporal, multisensor, and multiscale images. This is achieved using a convolution layer (conv3 $\times$ 3) with a kernel size of 3, followed by channel concatenation (Concat). This process facilitates the integration of spatial and spectral features from different levels, exemplified by merging H_1 with L_1 and H_3 with L_3. The multiscale features are combined through element-wise addition, resulting in $I = I_1 + I_3 + I_{\text{low}} + I_{\text{high}}$. Here, I_1 and I_3 represent the FF of H_1 with L_1 and H_3

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with L_3, respectively, while I_{low} and I_{high} represent the FF of all low-resolution and high-resolution images, respectively. This process ultimately transforms the original image into a high-dimensional feature space defined by $I \in \mathbb{R}^{H\times W\times N}$. Subsequently, the image is processed by an AFE block, which extracts spatial and frequency features, denoted as $I_0 \in \mathbb{R}^{H\times W\times N}$. For the encoder, the image I_0 pass through several STABs and Merging modules to generate the hierarchical feature representations. Specifically, the image I_0 then sequentially traverses N_1 STABs and a Merging module. This sequence is repeated with N_2 STABs, another merging stage. Following this, we design a symmetric transformer-based decoder. The decoder is composed of N_3 STABs, additional expanding layer, and finally N_4 STABs with a further Expanding module to enhance spatial and spectral feature extraction. The Merging modules employ MaxPooling 2×2 with a stride of 2, effectively reducing the feature map size by half and doubling the channel count. The feature at the ith encoder stage is denoted as $E_i \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 2^i N}$. In the expanding layer, each step involves upsampling the feature map, applying a conv 2×2 layer to reduce the channel count, and concatenating this with the corresponding cropped feature map from the contracting path, establishing skip connections to preserve border features. The feature at the ith stage of the decoder is $D_i \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 2^i N}$. The final reconstructed image, O, is obtained by summing I_0 and $O = I_0 + R$, followed by processing through a conv3 $\times$ 3 layer to produce the high-resolution prediction image $O' \in \mathbb{R}^{H\times W\times C}$. For practical application, we set N to 64 and configure STINet models with the combination (N_1, N_2, N_3, N_4) as (4, 4, 8, 8).

B. AFE Block

Time-series images, captured at various resolutions, contain an array of features across different scales. Notably, these features often show significant sparsity in the spectral dimension, posing a challenge for effective spectral correlation capture using convolutional methods in the initial stages of feature extraction [27], [31]. To address this issue, we propose the implementation of AFE, as illustrated in Fig. 1(b). This method is tailored to adaptively filter multilevel and multiscale information, aiming to extract relevant features efficiently [32]. In Sections II-B1 and II-B2, we provide an in-depth analysis of AFE, which includes two key components: AFF and FIE.

1) AFF: We incorporate pixel-adaptive convolution (PAC) [33] to effectively amalgamate features across various scales and accommodate variations both at the image and pixel levels. PAC introduces a spatially variable convolution kernel, symbolized as $K \in \mathbb{R}^{N'\times N\times S\times S}$, which significantly modifies the conventional spatially invariant convolution approach. This adjustment is crucial for the enhanced extraction of features that span multiple scales and levels. Specifically, for a feature map $I = (I_1, \ldots, I_n)$, where each component $I_i \in \mathbb{R}^N$ corresponds to a pixel covering $H \times W$ dimensions and comprises N channels. The convolutional process involves filter weights $W \in \mathbb{R}^{N'\times N\times S\times S}$, where N' and N respectively represent the number of output and input channels. This formulation allows the convolution operation to adapt to the spatial characteristics inherent in each pixel, thereby facilitating a more sophisticated and scale-sensitive extraction of features, essential for dealing with the complexities of spatial data in a multiscale context

$$I_i' = \sum_{j\in\Omega(i)} K\left(f_i, f_j\right) W\left[p_i - p_j\right] I_j + b \tag{1}$$

where K is the Gaussian: $K(f_i, f_j) = \exp(-(1/2)(f_i - f_j)^{\mathrm{T}}(f_i - f_j))$, $p_i = (x_i, y_i)^{\mathrm{T}}$ are pixel coordinates, $\Omega(\cdot)$ defines an $S \times S$ convolution window, and $b \in \mathbb{R}^N$ denotes biases. It should be noted that f represents "adaptive features," which are deep features obtained through end-to-end learning methods.

2) FIE: In our network architecture, we have incorporated specific structures dedicated to FIE. These structures are designed to discern both high-frequency details and low-frequency background information across varying scales. To effectively extract low-frequency global features from the adaptive features, we utilized a Transformer model equipped with long-short range attention (LSRA) [34]. This model includes a parallel branch specifically for capturing local dependencies, thereby enabling the attention mechanism to focus more efficiently on the global context. The formulation of this structure is as follows:

$$I_L' = L\left(I'\right) \tag{2}$$

where I_L' represents the low-frequency features. Acknowledging the crucial role of edge and texture details in feature extraction for image reconstruction, our objective is to preserve maximum high-frequency detail information throughout the frequency extraction process. An invertible neural network (INN) is capable of mapping between two distributions and can reverse this mapping [35]. In this study, we employ an Invertible layer to transform the feature. After fusing spatiotemporal features, this transformation targets a high-frequency distribution to reconstruct temporal changes. This approach serves as an efficient strategy for lossless feature extraction and is particularly effective for extracting shallow features across different modalities. In the Invertible layer, the input feature map I^k is split into two parts: I_a^k and I_b^k, which are later passed to I_a^{k+1} and I_b^{k+1} by affine calculation. The forward process can be formulated as

$$I_a^k, I_b^k = \text{split}\left(I^k\right) \tag{3}$$

where split denotes the partition operation of the feature maps. The input $I^k \in \mathbb{R}^{H\times W\times C}$ is transformed into $I_a^k \in \mathbb{R}^{H\times W\times(N/4)}$ and $I_b^k \in \mathbb{R}^{H\times W\times(3N/4)}$

$$\begin{cases} I_a^{k+1} = I_a^k \odot \exp\left(f_1\left(I_b^k\right) + f_2\left(I_b^k\right)\right) \\ I_b^{k+1} = I_b^k \odot \exp\left(f_3\left(I_a^{k+1}\right) + f_4\left(I_a^{k+1}\right)\right) \end{cases} \tag{4}$$

where $\odot$ represents pixel-by-pixel multiplication and $f_i (i = 1, \ldots, 4)$ represent arbitrary functions, enhancing the transformation capability can be achieved by integrating multiple coupling layers within a deep CNN.

Finally, I_a^{k+1} and I_b^{k+1} are then concatenated along the channel dimension to obtain I^{k+1}, i.e.,

$$I^{k+1} = \text{Concat}\left(I_a^{k+1}, I_b^{k+1}\right). \tag{5}$$

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Once the forward process is complete, the reverse process can be obtained. In contrast to the forward process, the resulting I^{k+1} is first divided into I_a^{k+1} and I_b^{k+1}, that is,

$$I_a^{k+1}, I_b^{k+1} = \text{split}\left(I^{k+1}\right). \tag{6}$$

The forward addition and multiplication operations are inverted to subtraction and division in the inverse conversion, i.e.,

$$\begin{cases} I_a^k = \left(I_a^{k+1} - f_2\left(I_b^k\right)\right)/\left(\exp\left(f_1\left(I_b^k\right)\right)\right) \\ I_b^k = \left(I_b^{k+1} - f_4\left(I_a^{k+1}\right)\right)/\left(\exp\left(f_3\left(I_a^{k+1}\right)\right)\right). \end{cases} \tag{7}$$

The process for extracting high-frequency information, denoted as I'_H, is defined in the following manner:

$$I'_H = H\left(I'\right). \tag{8}$$

The outputs of the high-frequency and low-frequency feature maps are fed into Convolution-ReLU-Convolution blocks, denoted as $I'_L \in \mathbb{R}^{H\times W\times (N/2)}$ and $I'_H \in \mathbb{R}^{H\times W\times (N/2)}$ respectively. The working process of the FF module can be summarized as follows:

$$I_0 = \text{Sigmoid}\left(\text{Concat}\left\{I'_L, I'_H\right\}\right) \tag{9}$$

Here, I_0 refers to the output of the AFE block. The feature map I_0 is designed with values ranging from 0 to 1. Constraining feature values to a specific range is crucial for enhancing the quality of image fusion outcomes.

C. Spatiotemporal-Wise Multihead Self-Attention

The utilization of nonlocal self-similarity in high-resolution image reconstruction is increasingly recognized in DL. However, traditional CNN often faces challenges in effectively modeling spatial and channel features [36]. Our research aims to explore the effectiveness of Transformer models in dynamic image reconstruction, given their notable capability to handle nonlocal, long-range dependencies, a strength evident in a variety of visual tasks. Confronting the limitations of the original Transformer model, particularly its focus on spatial long-range dependencies, we recognize its inadequacy in addressing varying-resolution images and cross-scale feature extraction. To overcome this, we propose the integration of MSA, illustrated in Fig. 2. In our model, each spatiotemporal feature is considered a separate token, with self-attention mechanisms applied across spatiotemporal dimensions. The inclusion of spatiotemporal self-attention (STA) within the ST-MSA module emphasizes the dynamic interplay of spatial and temporal elements, enhancing the capture of detailed spatial and temporal variations in images of diverse resolutions.

From a layer-normalized tensor $I \in \mathbb{R}^{H\times W\times N}$, our MSA first generates Q, K, and V projections, enriched with local context

$$Q = P^Q(I), K = P^K(I), V = P^V(I) \tag{10}$$

where P^Q, P^K, and P^V are projection weights in the linear layer. The process begins by applying a conv1 $\times$ 1 layer to collate pixel-wise cross-channel context, followed by a conv3 $\times$ 3 layer to capture spatial context. This results in the generation of Q, K, and V as $Q = XW^Q$, $K = XW^K$, and $V = XW^V$, where W^Q, W^K and $W^V \in \mathbb{R}^{C\times C}$ are learnable

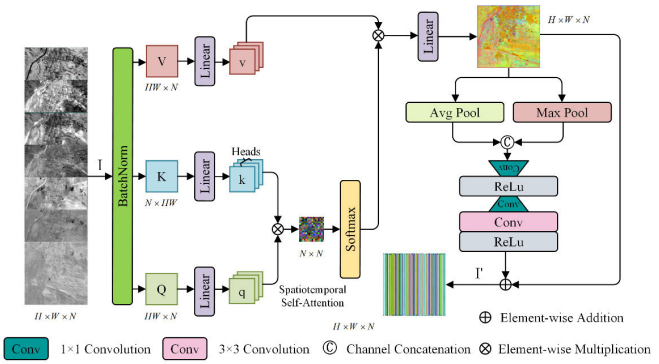


Fig. 2. Illustrates the key component embedded in STINet, which is the ST-MSA.

weights, and biases are excluded for simplicity. Following this, the queries, keys, and values are linearly projected H times to form $Q = [Q_1, \ldots, Q_H]$, $K = [K_1, \ldots, K_H]$, and $V = [V_1, \ldots, V_H]$. This setup enables the execution of self-attention across these multiple heads in parallel, thus facilitating the MSA mechanism

$$\text{head}'_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{11}$$

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d}}\right)V \tag{12}$$

where W_i^Q, W_i^K, and W_i^V present parameters in different projections and $(d)^{1/2}$ is a learning scaling parameter. Once the spatial-wise MSA results are computed, a temporal attention module is employed. This module processes the concatenated outputs of Max pooling and Average pooling along the channel axis. These concatenated outputs are then fed into convolution layers, designed to learn both shallow and deep features indicative of temporal changes.

III. Experimental Data and Processing

A. Study Site and Data

We are primarily concerned with resolving the issue of the continuous missing images by diverse land cover types and significant vegetation cover changes, multisensor inputs, and varying temporal spans. In this study, three scenarios were selected to evaluate the performance of our method. The simulated missing area is used as the test area, while the rest is the training area. In the training area, 80% of these patches were randomly selected for the training set, with the remaining patches used for the validation data The input data comprised coarse-resolution images, specifically MCD43A4 with a resolution of 500 m (L_1, L_2, L_3), supplemented by two sets of high-resolution images (H_1, H_3) from Landsat-8 (bands 1–7) or Sentinel-2 (bands 2–4 for blue, green, red; band 8 for near-infrared; and bands 11 and 12). The output generated by our model was a high-resolution predictive image (H_2), including seven bands from Landsat or six bands from Sentinel-2. All images, whether from MCD43A4, Landsat-8, or Sentinel-2, underwent resampling to a standardized spatial resolution of 30 or 10 m using the nearest neighbor interpolation method. Radiometric normalization was also applied to

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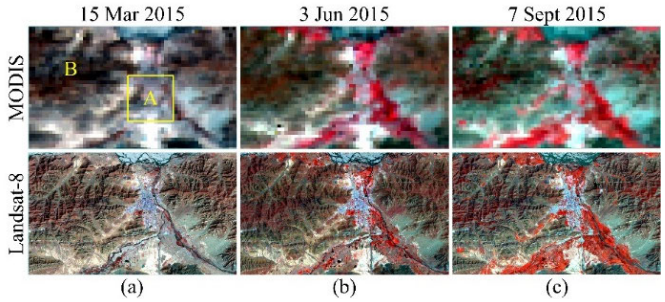


Fig. 3. False-color composite images (NIR, red, and green displayed as RGB) representing the training and test datasets for scenario 1, captured on (a) March 15, 2015, (b) June 3, 2015, and (c) September 7, 2015. A is the test region, B is the training region.

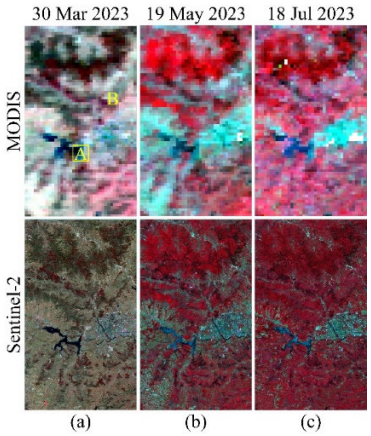


Fig. 4. False-color composite images (NIR, red, and green displayed as RGB) representing the training and test datasets for scenario 2, captured on (a) March 30, 2023, (b) May 19, 2023, and (c) July 18, 2023. A is the test region, B is the training region.

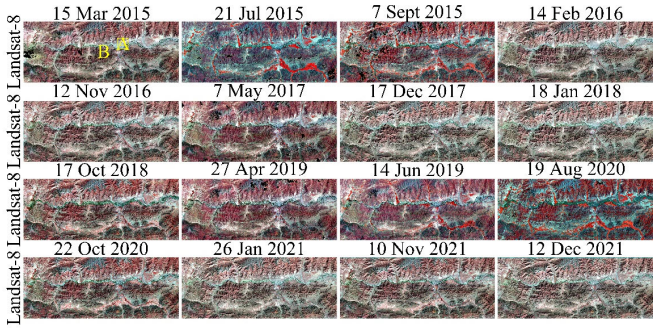


Fig. 5. False-color composite view (NIR, red and green as RGB) of the time-series training data and test data of scenario 3.

all datasets to minimize inconsistencies due to sensor-specific variations. This approach enabled a detailed examination of how data from different acquisition dates affect pixel-based image synthesis optimization and provided a robust validation of our method's effectiveness. Additionally, we performed NDVI calculations using reference images from both Landsat-8 and Sentinel-2, establishing a benchmark for comparison with other methods.

Scenario 1 is situated in Shigatse, situated in the southwestern region of Xizang, China (88°51′E, 29°15′N). This region is characterized by a diverse range of land cover types, including woodland, cropland, water area, and unused land. Notably, Shigatse has undergone significant land cover transformations, primarily influenced by infrastructural developments and agricultural expansion. The test area is region A (the simulated missing area), and the training area is B (not including A). To this research, we utilized three pairs of Landsat-8 and MCD43A4 images, captured on the following dates: March 15, 2015 (T1), June 3, 2015 (T2), and September 7, 2015 (T3). These images encompassed an area of approximately 18 × 30.6 km (600 × 1020 pixels), offering a spatial resolution of 30 m. The false color composites of these satellite images are presented in Fig. 3, providing a visual representation of the area under study.

Scenario 2 is situated in Jinan, in the central region of Shandong Province, Eastern China (118°6′E, 36°12′N). For this site, we acquired three pairs of Sentinel-2 and MCD43A4 images on specific dates: March 30, 2023 (T1), May 19, 2023 (T2), and July 18, 2023 (T3). These images span an area of approximately 3.4 × 2 km (3400 × 2000 pixels), with a spatial resolution of 10 m. As depicted in Fig. 4, the primary land cover types in this region include croplands, grasslands, and water areas. The entire image is divided into a training area (region B) and a test area (region A). Fig. 4(a)–(c) illustrates the significant spectral characteristic variations observed across these images, with a particular emphasis on aquatic and vegetative cover changes. Over the observed period, there was a noticeable increase in vegetation cover within the study area, including regrowth in regions previously affected by disturbances such as deforestation and infrastructure developments. The vegetative cover exhibited a gradual increase starting from March 30 and reaching its peak on July 18, indicative of substantial ecological changes in the region.

Scenario 3 is in Shigatse, once again. This location offers an expansive dataset, comprising 16 pairs of Landsat-8 and MCD43A4 images, collected from March 2015 to December 2021. These images cover a broad area measuring 45 × 120 km (1500 × 4000 pixels), offering a spatial resolution of 30 m. Fig. 5 illustrates the significant changes in land use and landscape that have occurred in this region, including agricultural development, infrastructural growth, and shifts in crop cultivation practices. The entire image is divided into a training area (region B) and a test area (region A).

The specific dates when these images were captured in all three study scenarios are systematically listed in Table I. This detailed chronology aids in a comprehensive analysis of the long-term environmental and land-use changes in the three areas.

B. Experimental Design

To ascertain the effectiveness of STINet, it was benchmarked against four established methodologies: ESTARFM [11], FSDAF [13], ESRCNN [20], and MUSTFN [24]. ESTARFM and FSDAF represent traditional approaches based on geostatistical spatial–spectral interpolation, while ESRCNN and MUSTFN are advanced, DL-based methods. ESRCNN marked an early application of DL in remote sensing fusion, and MUSTFN is recognized for its proficiency in multiscale STF. The training and development of these algorithms were performed on a

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TABLE I
ACQUISITION DATES OF LANDSAT-MODIS DATA, SENTINEL-MODIS DATA, AND LANDSAT-MODIS TIME SERIES DATA USED IN EXPERIMENTS

Image index	Scenario 1	Scenario 2	Scenario 3
1	15 Mar 2015	30 Mar 2023	15 Mar 2015
2	3 Jun 2015	19 May 2023	21 Jul 2015
3	7 Sept 2015	18 Jul 2023	7 Sept 2015
4			14 Feb 2016
5			12 Nov 2016
6			7 May 2017
7			17 Dec 2017
8			18 Jan 2018
9			17 Oct 2018
10			27 Apr 2019
11			14 Jun 2019
12			19 Aug 2020
13			22 Oct 2020
14			26 Jan 2021
15			10 Nov 2021
16			12 Dec 2021

specialized remote sensing dataset, tailored to optimize their respective performances. The training employed the Adam optimizer with an initial learning rate of 0.005 over a span of 500 epochs, reducing the learning rate by half every 50 epochs to ensure stable convergence. This training phase was focused on fine-tuning the hyperparameters of each model to augment their performance on the remote sensing dataset. STINet was developed using Python 3.9 and PyTorch 1.9 and tested on an Nvidia GeForce RTX 2080ti GPU with 64 GB RAM, facilitating efficient computation and accelerating the training process. The primary objective of the training was to minimize the root mean square error (RMSE) between the reconstructed and reference images, thereby enhancing the model accuracy and efficacy in remote sensing image reconstruction as follows:

1) *Evaluating STINet performance in diverse vegetation disturbance:* The objective of our experiment was to validate STINet's superior performance against five established STF methods. We concentrated on scenarios with similar or diverse vegetation covers, each marked by notable disturbances occurring within identical timeframes. Our chosen areas included a woodland undergoing infrastructural changes and two croplands influenced by irrigation and shifting cultivation practices. These varied disturbances provided a spectrum of scenarios showcasing vegetative changes. This experiment aims to showcase the efficacy of STINet in reconstructing the spatial distribution of disturbed vegetation cover, underlining its utility in precisely characterizing the dynamics of environmental transformations.
2) *Evaluating STINet performance in diverse vegetation recovery:* In this experiment, Sentinel-2 imagery was employed to depict vegetation recovery, facilitating the evaluation of STINet performance relative to other fusion methods. The Jinan study area, characterized by its diverse vegetation cover types, presented variability in vegetation growth patterns across different zones. This variability can be attributed to the range of growth conditions prevalent in each zone. Such a setting offered an optimal opportunity to assess STINet's capability in capturing and accurately representing the subtle aspects of vegetation recovery in heterogeneous landscapes, thus providing a comprehensive understanding of its applicability in diverse land cover changes.
3) *Predicting Vegetation Cover Changes in the Shigatse Region:* The primary challenge of this experiment was the prediction of vegetation cover disturbances and recovery, particularly in the context of limited comparable imagery. To address this, the focus was shifted to the Shigatse region to evaluate the efficacy of STINet in monitoring land cover changes. We utilized a comprehensive dataset comprising all available Landsat-MODIS image pairs, selecting key dates spanning from July 21, 2015, to November 10, 2021, for predictive analysis. During the training phase, the models ESRCNN, MUSTFN, and STINet were provided with sequences of three consecutive Landsat-MODIS image pairs. For the prediction tasks, the FSDAF model utilized the image pair obtained immediately prior to each prediction date, while ESTARFM additionally incorporated an image pair from just after the prediction date. The input datasets for ESRCNN and MUSTFN were aligned with those used for STINet, ensuring consistency in data processing. The objective of this experiment was to meticulously evaluate the proficiency of a singular model in detecting and depicting the dynamic changes in land cover over multiple time periods.

C. Evaluation Metrics

In our study, false color composites and error maps were employed for the qualitative analysis of STF results. For quantitative assessment, we utilized six established metrics: RMSE, dimensionless global relative error of synthesis (ERGAS), spectral angle mapper (SAM), universal image quality index (UIQI) [37], peak signal-to-noise ratio (PSNR) [38], and structure similarity index (SSIM) [39]. Higher values of UIQI, PSNR, and SSIM indicate better image reconstruction quality, reflecting a greater similarity and accuracy compared to the original images. Conversely, lower values of RMSE, ERGAS, and SAM are indicative of superior reconstruction, signaling fewer errors and discrepancies. Error maps, generated by calculating pixel-level differences between reconstructed and reference images, visually illustrate spectral discrepancies. Darker regions in these maps suggest higher consistency with the original images, while brighter areas indicate larger variances or inaccuracies. Absolute error, representing the average spectral reflectance difference across bands, was also used to quantify discrepancies in spectral accuracy.

IV. RESULT

A. Vegetation Disturbance Reconstruction (Experiment I)

In the quantitative analysis presented in Table II, STINet emerged as a superior performer, achieving the lowest values in RMSE, ERGAS, and SAM, and the highest in UIQI, PSNR, and SSIM, thereby outperforming other fusion methods.

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TABLE II

QUANTITATIVE RESULTS IN SCENARIO 1. RESULTS IN BOLD ARE THE BEST AND THOSE UNDERLINED ARE THE SECOND BEST

Model	RMSE	ERGAS	SAM	UIQI	PSNR	SSIM
ESTARFM	8.9271	17.9268	4.8286	0.7556	29.9699	0.8241
FSDAF	10.5850	24.6338	7.0557	0.6009	28.3489	0.7468
ESRCNN	7.1905	13.6937	<u>3.8783</u>	0.7613	31.9161	0.8195
MUSTFN	<u>6.7209</u>	<u>13.0007</u>	4.0494	<u>0.8056</u>	<u>32.5229</u>	<u>0.8339</u>
STINet	**5.0558**	**9.8587**	**3.3857**	**0.9095**	**34.9191**	**0.9069**

STINet demonstrated remarkable stability and precision, making it the most accurate among the compared methods. While ESTARFM exhibited slightly better performance and stability over FSDAF, the CNN-based methods, particularly due to their focus on spatial information extraction, showed significant improvements in maintaining spatial details. The integration of Transformers and CNN in STINet resulted in exceptional spatial reconstruction capabilities. This indicates that its spatial upsampling technique effectively captures both local and broader interactions, thereby enhancing the retention of spatial details. All models produced generally smoother fused images, but STINet, along with other CNN-based methods, excelled in surpassing conventional techniques. Significantly, STINet's performance surpassed that of both ESRCNN and MUSTFN, underscoring its effectiveness in complex spatial data fusion.

In Fig. 6, STINet demonstrated a distinct advantage in prediction accuracy, outperforming other fusion methods. In contrast, ESTARFM and FSDAF exhibited noticeable color discrepancies compared to reference images, particularly evident in area A as illustrated in Fig. 6(c)–(e). FSDAF results were compromised by spectral distortions, leading to color inaccuracies in certain regions. In scenarios like forest logging, agricultural irrigation, and shifting cultivation, especially under the considerable temporal changes present in the test data, both ESTARFM and FSDAF struggled to accurately reconstruct spatial details. Conversely, CNN-based methods ESRCNN and MUSTFN surpassed traditional techniques in spatial restoration. However, ESRCNN encountered issues with image blurring and spectral inaccuracies in specific areas, such as area B. MUSTFN, while showing enhanced spatial texture details across different land cover types, revealed clear spatial discrepancies from the reference image in regions with forest disturbances and shifting agriculture, as evident in area C. This limitation in MUSTFN might be attributed to its reliance on supervised learning and its limited adaptability to unrepresented vegetation disturbance scenarios in the training dataset. STINet fusion results are closely aligned with the reference images, surpassing the capabilities of the other four methods. This superior performance of STINet is primarily due to its spatiotemporal-wise transformer effectiveness in learning a wide array of spectral–spatial features, enabling it to adapt effectively to temporal changes in the prediction data. This attribute allows STINet to accurately capture the dynamics of target areas, resulting in highly reliable and precise image fusion outputs.

Fig. 7 presents error maps with values ranging from 0 to 1, offering a direct visual comparison between the reconstruction results of various STF methods and the reference image. The consistency between these error maps and the quantitative evaluation data is readily apparent. Notably, STINet outperforms all CNN-based models in this analysis, with MUSTFN following closely in terms of performance. MUSTFN's capability to retain spatial features is evident from the reduced spatial errors and anomalies in its error maps. However, in areas experiencing rapid land cover changes, other models demonstrate increased spectral errors, highlighting their limitations in accurately capturing land cover dynamics. A significant observation is the higher error rates for water areas compared to grassland and unused land, indicating a weaker responsiveness of these models to seasonal variations, particularly in riverine areas. This discrepancy is largely attributed to the limited representation of human-induced disturbances in the training dataset. In terms of reflectance value comparisons, STINet produces spectra closely matching the reference image, indicating its effectiveness in generating accurate reconstructions. This ability of STINet emphasizes its advanced performance in managing complex spatial and

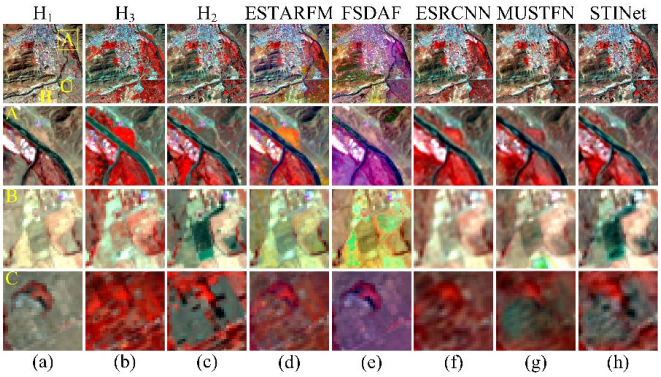


Fig. 6. Presents the outcomes of fusion Landsat-8 and MODIS data in scenario 1 using different methods. (a) and (b) Represent the starting and ending of the fusion process, respectively. (c) Displays the ground truth data. Panels (d)–(h) depict the results obtained using the ESTARFM, FSDAF, ESRCNN, MUSTFN, and STINet models, respectively. The images are rendered in a false-color composite view, utilizing near-infrared, red, and green channels as RGB.

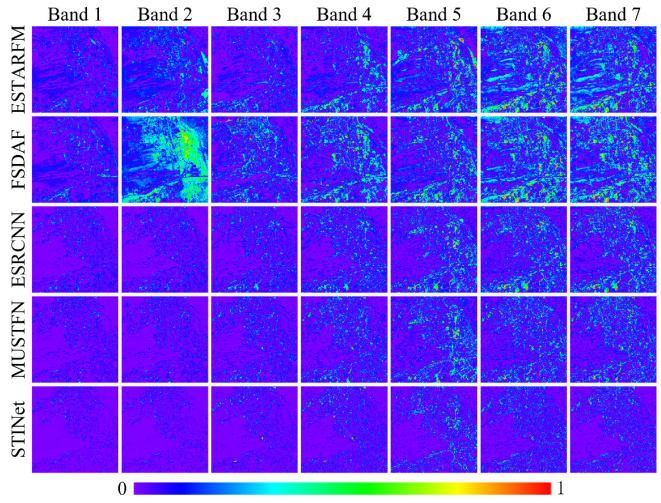


Fig. 7. Visual results of error maps in scenario 1. From the top to the bottom: error maps generated by ESTARFM, FSDAF, ESRCNN, MUSTFN, and by the proposed STINet.

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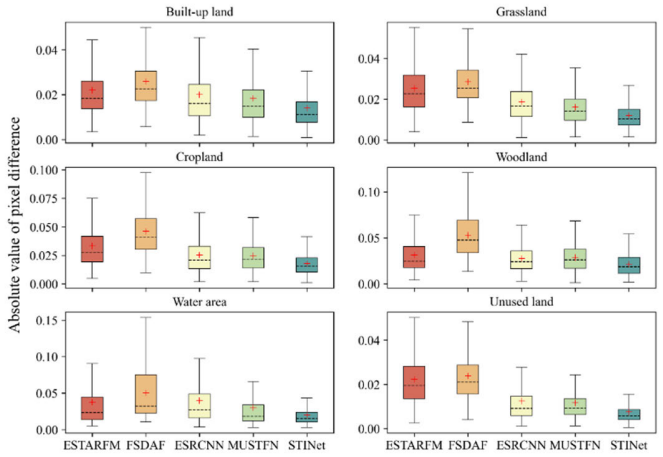


Fig. 8. Comparative analysis of different methods' performance in absolute errors across various land cover types.

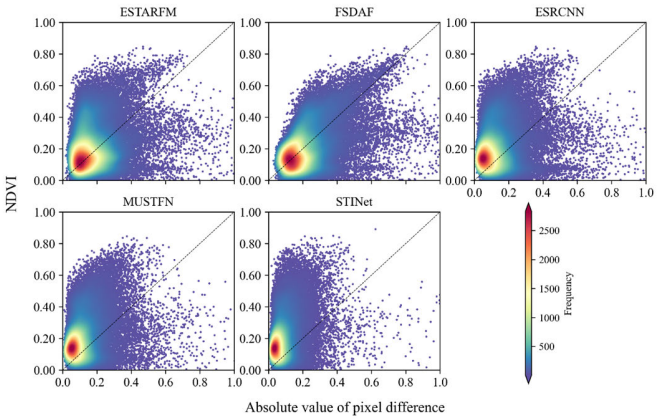


Fig. 9. Figure displays scatterplots that compare NDVI values from a reference image against the absolute errors of five different methods. The intensity of the color indicates point density, with warmer hues denoting higher concentrations and the black line is 1:1 line.

spectral variations, distinguishing it from other models in terms of accuracy and consistency in image reconstruction.

Fig. 8 illustrates a boxplot analysis of absolute pixel differences conducted by various STF models, encompassing multiple land cover types. In the statistical analysis, an equal number of pixels were randomly selected from each land cover type for comparison. Cropland and woodland areas show notably higher absolute errors compared to other land types. Among the methods analyzed, deep-learning-based approaches specifically ESRCNN, MUSTFN, and STINet outperform others, with STINet achieving the lowest absolute pixel differences across all land cover types. This is followed by the weight function-based method (ESTARFM), and finally, the unmixing-based method (FSDAF). Notably, STINet also demonstrates the smallest range of absolute errors, particularly in areas of lower unused land with vegetation cover. These findings indicate that traditional STF methods might encounter increased uncertainty in regions with higher vegetation cover.

Fig. 9 presents scatterplots that contrast NDVI values from a reference image with the Absolute errors of five distinct fusion methods. Notably, the scatterplot for STINet displays a highly concentrated distribution with narrow, Y-axis-aligned scattering stripes, indicating its superior performance over the other methods. This pattern reflects STINet's heightened spectral accuracy, especially in regions characterized by dense vegetation. These scatterplots effectively illustrate STINet's proficiency in delivering accurate NDVI measurements, emphasizing its capabilities in precise STF and accurately mirroring vegetation conditions.

B. Vegetation Recovery Reconstruction (Experiment II)

Table III presents a quantitative evaluation of the fusion results obtained from integrating MODIS and Sentinel-2 imagery. This assessment indicates that CNN-based methods, particularly STINet, demonstrate significant advantages over traditional fusion methods, with STINet emerging as the most effective. Comparing these results with the quantitative data from the Shigatse dataset reveals a general increase in fusion accuracy across six indices for all STF methods. This enhancement in performance can be attributed to STINet's capability to utilize inherent spatiotemporal-spectral change features in high-resolution imagery, effectively maintaining spectral accuracy while enhancing spatial resolution.

TABLE III

QUANTITATIVE RESULTS IN SCENARIO 2. RESULTS IN BOLD ARE THE BEST AND THOSE UNDERLINED ARE THE SECOND BEST

Model	RMSE	ERGAS	SAM	UIQI	PSNR	SSIM
ESTARFM	9.0446	13.2826	5.4418	0.7359	29.7616	0.8640
FSDAF	16.5695	25.6545	11.3554	0.5871	24.6060	0.8003
ESRCNN	6.6628	10.1075	3.3439	0.7978	32.1983	0.8663
MUSTFN	6.4948	9.5764	3.5864	0.8355	32.7221	0.8779
STINet	**4.6235**	**6.9635**	**2.5513**	**0.9249**	**35.3592**	**0.9164**

Fig. 10 effectively illustrates the performance of various STF methods across different land cover types, including cropland (Region A), woodland (Region B), and built-up land (Region C). The figure clearly demonstrates the variability in fusion outcomes among the five methods. Specifically, ESTARFM results in agricultural and aquatic environments, as shown in Fig. 10(d), lack precision in spatial feature restoration. FSDAF, on the other hand, exhibits notable spectral discrepancies, with the colors in its zoomed areas appearing lighter than the reference image, as observed in Fig. 10(e). While ESRCNN shows a relative advantage in restoring spatial details over these methods, it still presents some blurring and spectral inconsistencies in certain areas, as depicted in Fig. 10(f). MUSTFN, although superior in both spatial and spectral dimensions compared to the first three methods, tends to lack texture in its results, as evident in Fig. 10(g). In contrast, STINet aligns closely with the reference images, demonstrating its proficiency in predicting abrupt land cover changes and reconstructing complex spatial structures. This comparison underscores STINet's advanced capabilities in spatiotemporal data fusion, particularly in accurately rendering complex land cover changes.

Fig. 11 features error maps that assess the discrepancies between the results of various fusion methods and reference images in a study area predominantly comprising cropland, woodland, and built-up land. Compared to natural vegetation, the spatial and spectral properties of constructive lands exhibit less variability over time, making them more stable.

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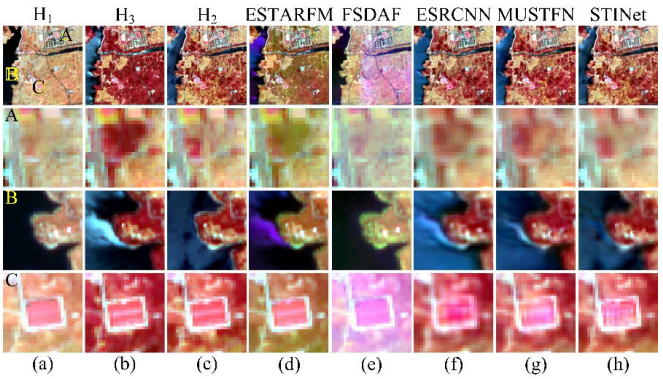


Fig. 10. Presents the outcomes of fusion Sentinel-2 and MODIS data in scenario 2 using different methods. (a) and (b) Represent the starting and ending of the fusion process, respectively. (c) Displays the reference image data. Panels (d)–(h) depict the results obtained using the ESTARFM, FSDAF, ESRCNN, MUSTFN, and STINet models, respectively. The images are rendered in a false-color composite view, utilizing near-infrared, red, and green channels as RGB.

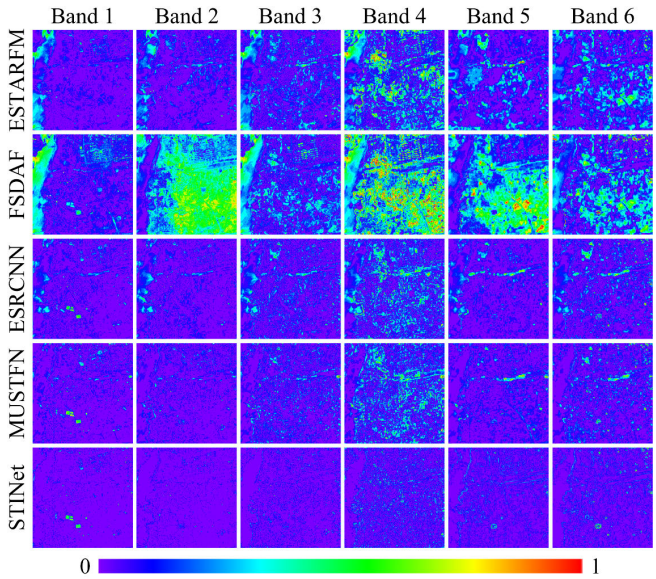


Fig. 11. Visual results of error maps in scenario 2. From the top to the bottom: error maps generated by ESTARFM, FSDAF, ESRCNN, MUSTFN, and by the proposed STINet.

In contrast, vegetation types like grasses and trees undergo significant seasonal changes, thereby impacting the sensitivity of these structures to temporal alterations. The performance of supervised learning-based methods tends to be less optimal when the vegetation recovery conditions in the test data significantly differ from those in the training data. However, STINet excels in restoring the integrity of building structures and capturing coarse textures, as visually confirmed through error map comparisons. Notably, the error maps for ESTARFM, FSDAF, and ESRCNN show considerable blurriness near riverbanks, likely due to registration issues leading to spectral distortions. Specifically, FSDAF exhibits pronounced inaccuracies in bands 2, 4, and 5, potentially due to its limited capability in reconstructing vegetation changes.

Fig. 12 presents boxplots illustrating the absolute pixel differences for five distinct STF methodologies under scenario 2, with a notable performance superiority of the STINet

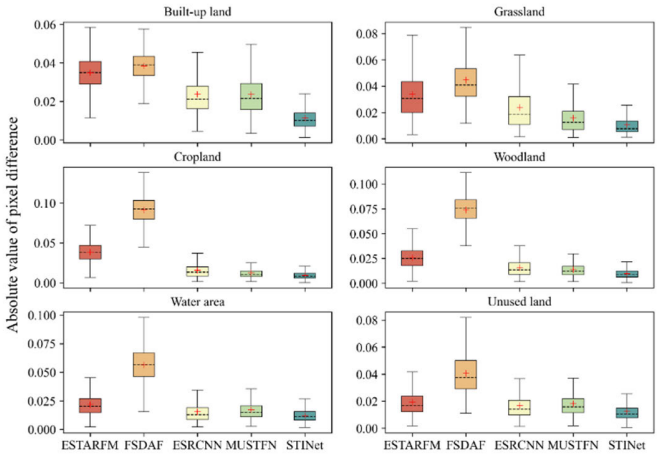


Fig. 12. Comparative analysis of different methods' performance in absolute errors across various land cover types.

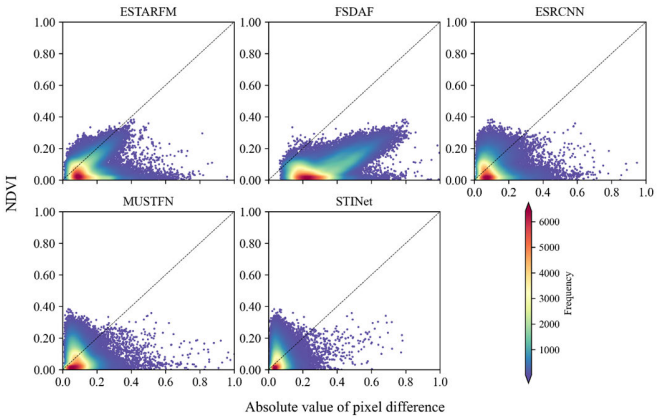


Fig. 13. Figure displays scatterplots that compare NDVI values from a reference image against the absolute errors of five different methods. The intensity of the color indicates point density, with warmer hues denoting higher concentrations and the black line is 1:1 line.

across various land cover types. This observation underscores STINet's capability to maintain low uncertainty levels while ensuring high precision outcomes, particularly in regions characterized by dense vegetation. Concurrently, Fig. 13 employs scatterplots to compare the absolute errors of the fusion methods against the NDVI values derived from a reference image. The scatterplots utilize a color gradient to represent point density visually, where warmer hues signify areas of higher density. Among the methods evaluated, STINet distinguishes itself with a tightly concentrated scatter distribution, indicative of the lowest absolute errors. The consolidated data from Figs. 12 and 13 corroborate STINet's exemplary accuracy in predicting NDVI values, affirming its unparalleled precision across all spectral bands. This evidence suggests that STINet is highly effective in spectral analysis, offering significant advantages for applications requiring accurate land cover and vegetation assessments.

C. Time-Series Reconstruction (Experiment III)

To demonstrate STINet's effectiveness in reconstructing time-series images, particularly in scenarios involving ongoing disturbances and recoveries, we conducted an accuracy

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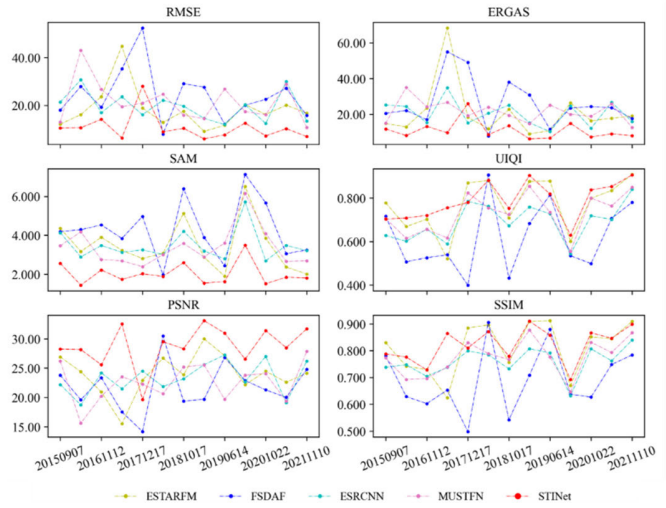


Fig. 14. Quantitative accuracy assessment of the fusion results in Experiment III. Metrics are averaged over all the seven bands.

assessment of five STF methods. As depicted in Fig. 14, STINet consistently outperforms other methods in accuracy across various dates. The figure showcases fusion results with Landsat-8 images at the respective prediction dates serving as benchmarks (see Fig. 15). It is of particular interest that the reconstructions for the dates of December 17, 2017, January 18, 2018, and June 14, 2019, demonstrated suboptimal performance in accurately representing diverse land use scenarios such as deforestation, flood events, and urban sprawl. Vegetation is highly sensitive to temporal changes, especially to disturbances and recovery processes, leading to significant differences between consecutive remote sensing images. In undisturbed high vegetation cover areas, ESTARFM and FSDAF perform best, as shown in the magnified sections of Fig. 16(d) and (e). The fusion results of ESRCNN and MUSTFN appear overly smooth in the magnified sections, lacking detailed spatial resolution, as depicted in Fig. 16(f) and (g). STINet outperforms these methods in visual quality, yet it falls short in accurately reflecting drastic changes in areas with low vegetation cover, particularly in flood zones explicitly present in the reference data.

Fig. 17 utilizes boxplots to compare the absolute errors of various methods across different land cover types. The observations indicate that traditional methods underperform in accurately simulating the spectral reflectance for various land types, especially in areas with high vegetation cover. In contrast, STINet demonstrates significantly lower errors compared to other methods, highlighting its substantial advantage in enhancing the precision of reflectance reconstruction.

Fig. 18 presents scatter density plots, in which STINet scatter stripes are noticeably more condensed compared to those of other methods, indicating its superior predictive accuracy. In contrast, the scatter stripes of the two traditional STF methods are broader than those of ESRCNN and MUSTFN. The broadening of scatter stripes illustrates significant discrepancies in vegetation cover between actual and predicted data. This observation suggests that traditional methods may face challenges in accurately predicting over larger temporal gaps in vegetation cover. In comparison, STINet demonstrates consistently high accuracy across varying time intervals, highlighting its robustness in handling temporal variations in vegetation cover.

Fig. 19 presents the NDVI time series extracted from 14 fused Landsat images, illustrating significant vegetation cover discrepancies between prediction and ground truth. This disparity leads to notable differences in the fusion results of all five methods compared to the reference data. In the reconstruction conducted using the ESTARFM, ESRCNN, and MUSTFN methods, the NDVI time series for unused land demonstrated higher fidelity to the actual observed values. Conversely, in woodland areas characterized by dense vegetation, significant deviations were observed between the model outputs and the reference data. This discrepancy may be attributed to seasonal and interannual variations in vegetation cover on auxiliary dates and their differing sensitivities to the predicted data. Similarly, FSDAF shows deviations from the reference values in specific areas, particularly in river valleys, as depicted in cropland. In grassland areas, the MUSTFN results also display higher errors. Overall, the STINet method provides more accurate predictions in regions with significant vegetation coverage changes between auxiliary and prediction dates, where STINet results closely resemble the reference data.

V. Discussion

A. STF in Phenological Changes

Several STF methods have been developed to enhance the spatial resolution of Sentinel-2 imagery, primarily leveraging auxiliary data from lower-resolution sources like MODIS. These methods fall into two categories: the first relies on spatial similarity, assuming that neighboring pixels with similar characteristics possess comparable spectral profiles, as exemplified by ESTARFM. The second category depends on temporal autocorrelation in MODIS data, using spatial interpolation to improve resolution, such as in the FSDAF approach. Traditional data fusion techniques typically depend on pairing fine and coarse-resolution images for each prediction cycle. In contrast, DL-based methods like ESRCNN and MUSTFN excel in extracting features and learning nonlinear relationships between high and low-resolution images. However, these methods may exhibit instability when training and testing datasets have significant differences. STINet distinguishes itself by integrating spatial–spectral features and leveraging temporal and spatial change detection to enhance the use of time-varying information. Our experimental results and analyses demonstrate that STINet outperforms other methods (ESTARFM, FSDAF, ESRCNN, and MUSTFN), particularly in scenarios with diverse vegetation cover changes, such as disturbance and recovery in vegetation dynamics. STINet uses time-series image pairs as inputs, effectively fusing spatial–spectral features to closely approximate the target image. Notably, STINet performance consistently improves with the inclusion of more input time-series image pairs, handling the fusion process efficiently without the need for explicit change prediction.

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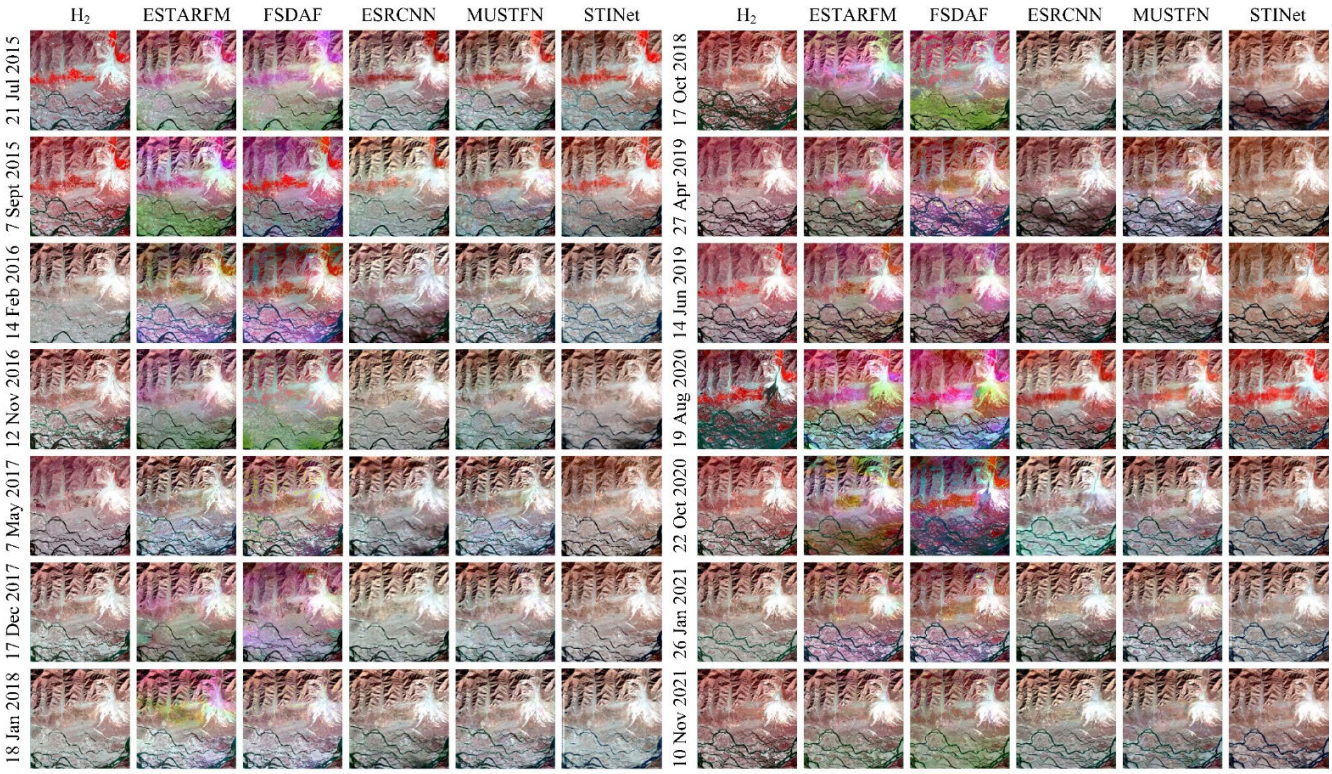


Fig. 15. Presents the outcomes of fusion Landsat-8 and MODIS data in Experiment III using different methods. The images are rendered in a false-color composite view, utilizing near-infrared, red, and green channels as RGB.

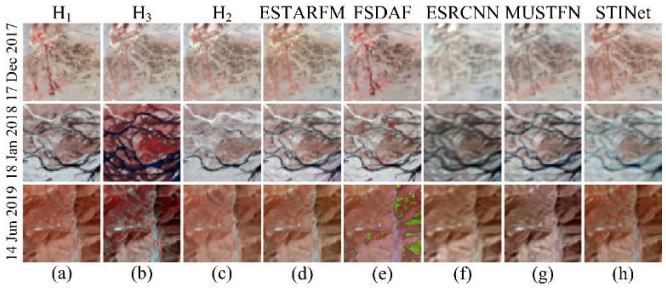


Fig. 16. Presents the outcomes of fusion Landsat-8 and MODIS data in Experiment III using different methods. (a) and (b) Represent the starting and ending of the fusion process, respectively. (c) Displays the reference image data. Panels (d)–(h) depict the results obtained using the ESTARFM, FSDAF, ESRCNN, MUSTFN, and STINet models, respectively. The images are rendered in a false-color composite view, utilizing near-infrared, red, and green channels as RGB. Each line represents woodland, bareland, and grassland, respectively.

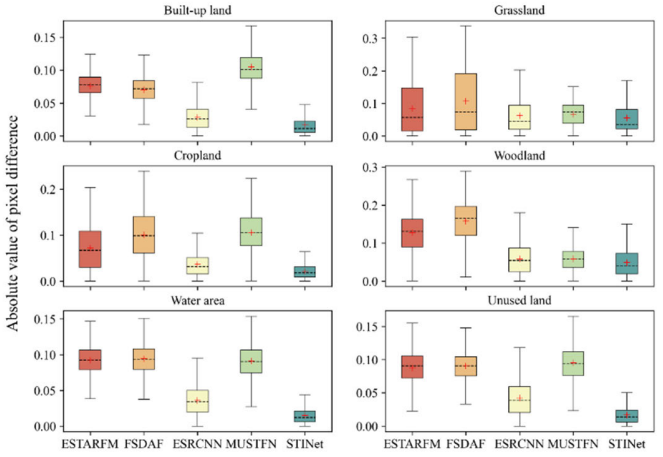


Fig. 17. Comparative analysis of different methods' performance in absolute errors across various land cover types.

Fig. 20 illustrates the challenges associated with asynchronous phenological changes between input data and predicted data in remote sensing. In this study, we utilized pairs of time-series images to predict Sentinel-2 imagery within the northeastern region of Wuwei, China (103°13′E, 38°8′N). The images reconstructed by STINet demonstrate enhanced accuracy in spatial details, offering a more precise depiction of the actual ground conditions. This improvement is indicative of STINet's ability to effectively handle phenological discrepancies between the time-series input data and the ground truth.

The enhancement in accuracy is not only visually evident but also quantitatively substantiated through established accuracy evaluation metrics, as depicted in Fig. 21. While ESTARFM and FSDAF perform badly in quantitative assessments on most prediction dates, they exhibit inferior visual performance compared to CNN-based fusion models. These traditional STF methods predict using images from adjacent time points, making it challenging to accurately capture unexpected drastic changes in noncontinuous and unstable time series fluctuations. In contrast, CNN-based STF methods reconstruct land cover change patterns over multiple temporal phases, enabling them to predict a broader range of variations. This approach results in more diverse and accurate visual

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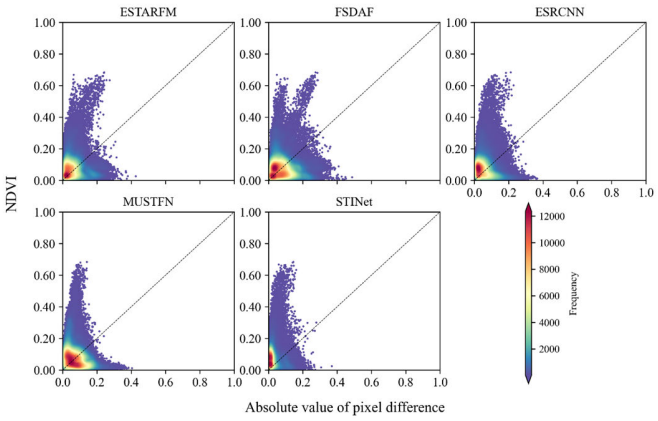


Fig. 18. Figure displays scatterplots that compare NDVI values from a reference image against the absolute errors of five different methods. The intensity of the color indicates point density, with warmer hues denoting higher concentrations and the black line is 1:1 line.

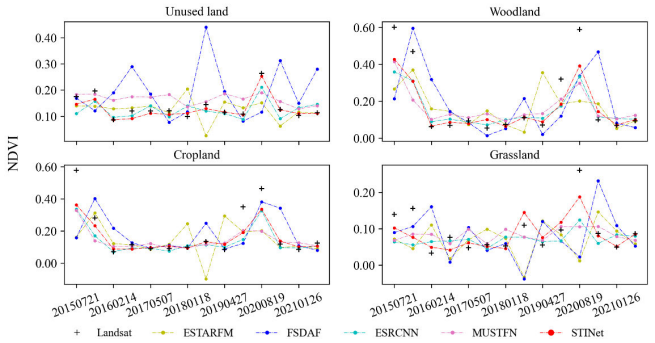


Fig. 19. Reconstructed NDVI images from various methods.

representations, particularly in scenarios involving significant land cover changes.

Further analysis substantiates STINet's superiority in accurately depicting temporal variations in land cover, validated by comparing NDVI time series extracted from fused reflectance images. To assess the capability of STINet in capturing phenological features of croplands from Sentinel-2 time series data, this study analyzed four strategically selected agricultural parcels. STINet demonstrated exceptional proficiency in identifying unique agricultural patterns in areas of double-cropping in Fig. 22 (Parcel A), a task that poses significant challenges to methods like ESTARFM and FSDAF, which are reliant on simulating complex temporal variations. As evidenced in Fig. 22 (Parcel B), STINet also proved adept at accurately reconstructing the NDVI for individual growing seasons. However, its predictive accuracy declined in regions with minor vegetation cover changes, as indicated by significant fluctuations in the reconstructed NDVI time series, likely due to its increased sensitivity to adjacent temporal vegetation variations in Fig. 22 (Parcel C, D). This adaptability underscores STINet's ability to precisely reconstruct NDVI curves for single growth seasons across diverse regions. The findings affirm STINet's superior capacity to learn phenological features across different crop types, showcasing its distinct advantage in identifying varied croplands within heterogeneous landscapes.

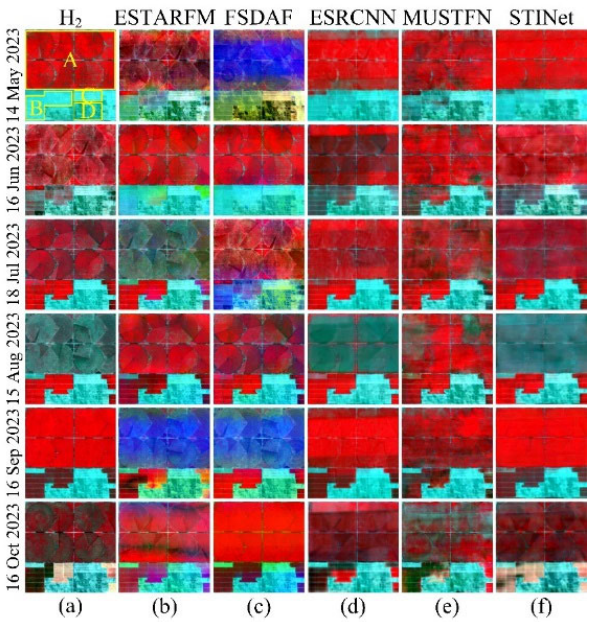


Fig. 20. Presents the outcomes of fusion Sentinel-2 and MODIS data using different methods. (a) Displays the reference image data. Panels (b)–(f) depict the results obtained using the ESTARFM, FSDAF, ESRCNN, MUSTFN, and STINet models, respectively. The images are rendered in a false-color composite view, utilizing near-infrared, red, and green channels as RGB. Parcels A–D correspond to distinct agricultural practices. Specifically, Parcel A is associated with multiple cropping patterns within a single year, exceeding two cycles, Parcel B with a single annual cropping cycle, and Parcels C and D are not cultivated. However, the vegetation cover on Parcels C and D exhibits subtle variations, indicating differences in vegetation density despite the absence of cultivation.

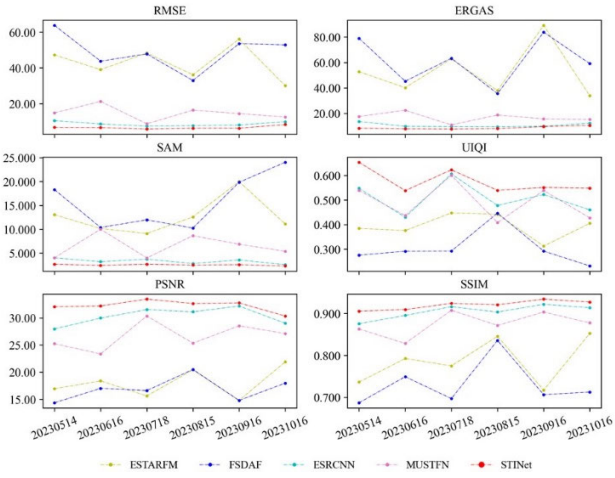


Fig. 21. Quantitative accuracy assessment of the fusion results. Metrics are averaged over all the six bands.

B. Ablation Study

In this section, ablation studies were conducted to evaluate the contributions of the FF block, AFE block, and ST-MSA. The detailed results of the data comparison analysis from Experiment I are systematically displayed in Table IV, with the corresponding visual outcomes illustrated in Fig. 23. The incorporation of the FF block enabled Model 5 to achieve a PSNR improvement of 0.7953 dB over the Baseline model, underscoring the efficiency and adaptability of our approach in integrating features across different sensors and scales.

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TABLE IV

RESULTS OF OUR ABLATION STUDY, PRESENTING THE PERFORMANCE METRICS OF DIFFERENT CONFIGURATIONS WITHIN THE STINET MODEL. THE MOST EFFECTIVE RESULTS ARE EMPHASIZED IN BOLD, WITH THE SECOND-BEST OUTCOMES UNDERLINED

Model	FF	AFE	ST-MSA	RMSE	ERGAS	SAM	UIQI	PSNR	SSIM
Baseline	✗	✗	✗	6.1950	11.9589	3.8993	0.8599	33.3228	0.8670
Model 1	✗	✓	✓	5.2049	<u>10.0949</u>	3.5638	<u>0.9046</u>	<u>34.7546</u>	<u>0.9029</u>
Model 2	✓	✗	✓	6.9466	13.6903	4.0648	0.8296	32.1269	0.8452
Model 3	✓	✓	✗	5.4141	10.5623	3.6118	0.8973	34.3432	0.8949
Model 4	✗	✗	✓	<u>5.3945</u>	10.5516	3.4075	0.8872	34.3719	0.8859
Model 5	✓	✗	✗	5.4018	10.9526	**3.3252**	0.8965	34.1181	0.8918
Model 6	✗	✓	✗	5.7537	11.3512	3.8970	0.8891	33.7425	0.8900
STINet	✓	✓	✓	**5.0558**	**9.8587**	<u>3.3857</u>	**0.9095**	**34.9191**	**0.9069**

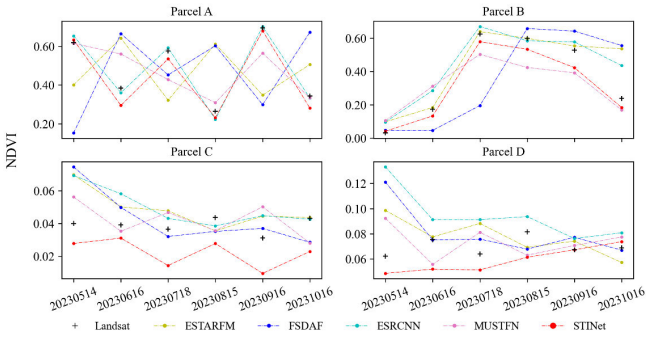


Fig. 22. Reconstructed NDVI images from various methods.

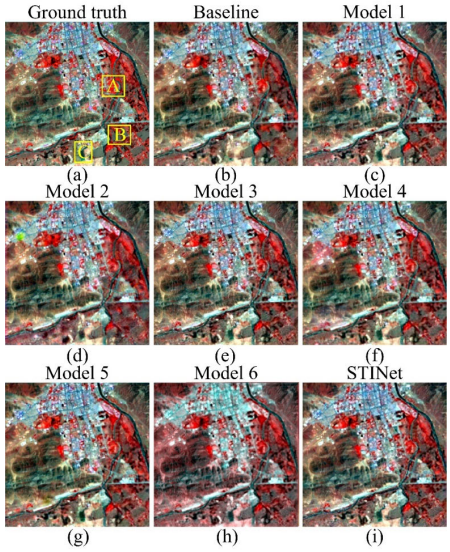


Fig. 23. Visual results from the ablation study. (a) Reference image. (b)–(i) Fused results under different configurations.

This module proficiently fuses multiscale spatial features over varying temporal phases. Furthermore, a comparison between Models 1, 4, and 6 illustrates that the inclusion of the AFE block and ST-MSA module within a U-shaped architecture network significantly enhances the quality of predicted images in scenarios of abrupt land cover changes and heterogeneous regions. To elucidate the impact of these two components on the overall model performance, they were methodically removed and analyzed in this section.

Our study focuses on the AFE block in STINet, designed to harness multiscale spatiotemporal information from remote sensing imagery to enhance fusion performance. Utilizing adaptive filtering, the AFE block extracts both low-frequency and high-frequency information, integral for incorporating a broad range of spectral features. Our findings, particularly in Region B as depicted in Fig. 23(d) and (h), demonstrate that the AFE block effectively elevates lower-level change characteristics to more prominent levels, enabling accurate reconstruction of complex vegetation disturbances. This method simplifies the optimization process and facilitates more effective collaboration within DL models, showcasing the adaptability of the AFE block in feature extraction. Furthermore, the integration of the ST-MSA with the AFE block in STINet has led to peak performance in metrics such as PSNR and SSIM. Particularly in Region A, as shown in Fig. 23(a), (b), and (i), the incorporation of ST-MSA, tailored for the 3-D structure of multispectral remote sensing images, significantly enhances the model spatial change detection capabilities. The synergy between AFE and ST-MSA optimally extracts spatial and spectral change information while preserving intricate spatial details, potentially augmented by multilevel Transformers. The ST-MSA design in STINet aims to exploit multiscale spatial and temporal variances in remote sensing imagery. By methodically applying spatiotemporal attention, it blends high-dimensional structural alterations with low-dimensional textural features, thereby enhancing the model's overall performance in spatiotemporal image fusion tasks. This sophisticated approach to feature extraction and attention mechanisms highlights STINet's advanced capabilities in processing remote sensing data for accurate image reconstruction.

C. Balancing Temporal Dynamics and Spatial Heterogeneity in STINet

In our STF experiments, the time intervals between base and predicted data emerged as a critical factor. We experimented with intervals ranging from 3 to 9 in our fusion tests. Unlike traditional STF methods like ESTARFM and FSDAF, which rely on pixel-by-pixel prediction, STINet utilizes intervals between coarse and fine image pairs, enabling single-step prediction of time-series data. This strategy allows for an increased number of training samples, yielding more robust results than ESTARFM and FSDAF, particularly for adjacent image pairs. However, overly large intervals between image pairs may overlook the heterogeneity of the target area, potentially leading to inaccuracies in predictions. To address this issue, we evaluated the sensitivity of our method to varying spans of image pairs across four subregions in Experiment III,

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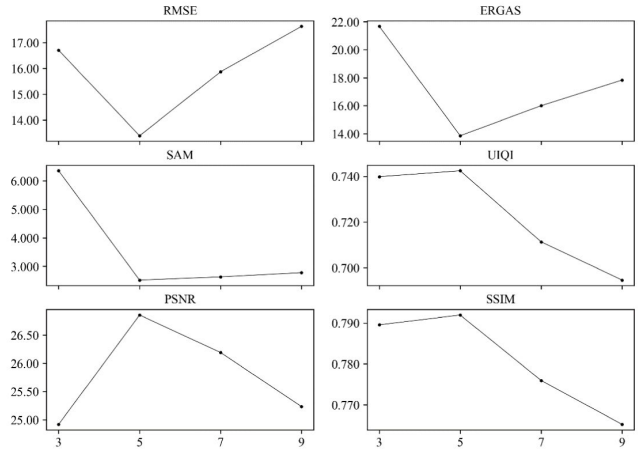


Fig. 24. Quantitative accuracy assessment of the fusion results. Established metrics are calculated as the average across 14 predicted images.

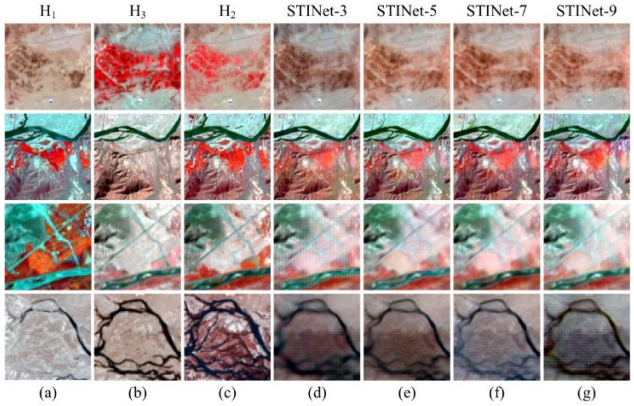


Fig. 25. Presents the outcomes of fusion Landsat-8 and MODIS data in Experiment III using different methods. (a) and (b) Represent the starting and ending of the fusion process, respectively. (c) Displays the reference image data. Panels (d)–(g) depict the results obtained using the STINet-3, STINet-5, STINet-7, and STINet-9 models, respectively. The images are rendered in a false-color composite view, utilizing near-infrared, red, and green channels as RGB. Each line represents woodland, bare land, grassland, and water area, respectively.

each characterized by different degrees of spatial heterogeneity. Figs. 24 and 25 illustrate the variation in established metrics as influenced by the temporal spans between image pairs and their corresponding predictions. As the temporal span increases, metrics such as RMSE, ERGAS, and SAM show a gradual decline, while the trends of UIQI, PSNR, and SSIM metrics exhibit an inverse relationship, underscoring the enhanced capability for extracting spatiotemporal change features. Notably, STINet demonstrates optimal performance at a temporal span of 5, indicating this as the ideal configuration for spatiotemporal analysis. Due to the increasing complexity and difficulty in training the model over larger temporal spans, a span of 5 is selected as the optimal setting. This configuration is utilized as a benchmark for evaluating the effectiveness of STINet in comparison to other methods.

VI. Conclusion

In this research, we developed STINet, an advanced spatiotemporal image fusion method, specifically engineered to tackle the complexities of multiscale and multisensor data fusion. STINet represents a substantial advancement over traditional image fusion methods, effectively harnessing the capabilities of CNNs to intricately extract and utilize key spatiotemporal dynamics. The model efficacy is demonstrated through comparative analysis of visual results, alongside assessments using six quantitative indices, ground truth reference imagery, and scatter density plots. STINet has proven its ability to accurately reconstruct time-series images, capturing both spatial and spectral variations across diverse land cover types. Overall, STINet is poised to make a significant impact on future remote sensing research and applications, offering a comprehensive and effective solution for spatiotemporal data challenges.

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References

[1] G. Forzieri, V. Dakos, N. G. McDowell, A. Ramdane, and A. Cescatti, "Emerging signals of declining forest resilience under climate change," *Nature*, vol. 608, no. 7923, pp. 534–539, Aug. 2022, doi: 10.1038/s41586-022-04959-9.

[2] R. E. Kennedy et al., "Attribution of disturbance change agent from Landsat time-series in support of habitat monitoring in the Puget Sound region, USA," *Remote Sens. Environ.*, vol. 166, pp. 271–285, Sep. 2015, doi: 10.1016/j.rse.2015.05.005.

[3] X. Wang et al., "Assessment of soil erosion change and its relationships with land use/cover change in China from the end of the 1980s to 2010," *Catena*, vol. 137, pp. 256–268, Feb. 2016, doi: 10.1016/j.catena.2015.10.004.

[4] J. Zhao, S. Huang, Q. Huang, H. Wang, G. Leng, and W. Fang, "Time-lagged response of vegetation dynamics to climatic and teleconnection factors," *Catena*, vol. 189, Jun. 2020, Art. no. 104474, doi: 10.1016/j.catena.2020.104474.

[5] R. R. Colditz, M. Schmidt, C. Conrad, M. C. Hansen, and S. Dech, "Land cover classification with coarse spatial resolution data to derive continuous and discrete maps for complex regions," *Remote Sens. Environ.*, vol. 115, no. 12, pp. 3264–3275, Dec. 2011, doi: 10.1016/j.rse.2011.07.010.

[6] D. Chu et al., "Long time-series NDVI reconstruction in cloud-prone regions via spatio-temporal tensor completion," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112632, doi: 10.1016/j.rse.2021.112632.

[7] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006, doi: 10.1109/TGRS.2006.872081.

[8] F. Meng et al., "Single remote sensing image super-resolution via a generative adversarial network with stratified dense sampling and chain training," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5400822, doi: 10.1109/TGRS.2023.3344112.

[9] Y. Yu, L. J. Renzullo, T. R. McVicar, B. P. Malone, and S. Tian, "Generating daily 100 m resolution land surface temperature estimates continentally using an unbiased spatiotemporal fusion approach," *Remote Sens. Environ.*, vol. 297, Nov. 2023, Art. no. 113784, doi: 10.1016/j.rse.2023.113784.

[10] T. Hilker et al., "A new data fusion model for high spatial- and temporal-resolution mapping of forest disturbance based on Landsat and MODIS," *Remote Sens. Environ.*, vol. 113, no. 8, pp. 1613–1627, Aug. 2009, doi: 10.1016/j.rse.2009.03.007.

[11] X. Zhu, J. Chen, F. Gao, X. Chen, and J. G. Masek, "An enhanced spatial and temporal adaptive reflectance fusion model for complex heterogeneous regions," *Remote Sens. Environ.*, vol. 114, no. 11, pp. 2610–2623, Nov. 2010, doi: 10.1016/j.rse.2010.05.032.

[12] B. Zhukov, D. Oertel, F. Lanzl, and G. Reinhäckel, "Unmixing-based multisensor multiresolution image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1212–1226, May 1999, doi: 10.1109/36.763276.

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[13] X. Zhu, E. H. Helmer, F. Gao, D. Liu, J. Chen, and M. A. Lefsky, "A flexible spatiotemporal method for fusing satellite images with different resolutions," *Remote Sens. Environ.*, vol. 172, pp. 165–177, Jan. 2016, doi: 10.1016/j.rse.2015.11.016.

[14] C. Xu, X. Du, Z. Yan, J. Zhu, S. Xu, and X. Fan, "VSDF: A variation-based spatiotemporal data fusion method," *Remote Sens. Environ.*, vol. 283, Dec. 2022, Art. no. 113309, doi: 10.1016/j.rse.2022.113309.

[15] Z. Gu, J. Chen, Y. Chen, Y. Qiu, X. Zhu, and X. Chen, "Agri-fuse: A novel spatiotemporal fusion method designed for agricultural scenarios with diverse phenological changes," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113874, doi: 10.1016/j.rse.2023.113874.

[16] S. M. Herrmann, A. Anyamba, and C. J. Tucker, "Recent trends in vegetation dynamics in the African Sahel and their relationship to climate," *Global Environ. Change*, vol. 15, no. 4, pp. 394–404, Dec. 2005, doi: 10.1016/j.gloenvcha.2005.08.004.

[17] A. N. Na et al., "An improved spatio-temporal fusion model for remote sensing images based on singular spectrum analysis," *Remote Sens. Natural Resour.*, vol. 35, no. 2, pp. 89–96, 2023, doi: 10.6046/zrzyyg.2022095.

[18] K. Knauer, U. Gessner, R. Fensholt, and C. Kuenzer, "An ESTARFM fusion framework for the generation of large-scale time series in cloud-prone and heterogeneous landscapes," *Remote Sens.*, vol. 8, no. 5, p. 425, May 2016, doi: 10.3390/rs8050425.

[19] C. Dong, C. C. Loy, K. He, and X. Tang, "Learning a deep convolutional network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, in Lecture Notes in Computer Science: Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics, vol. 8692, Sep. 2014, pp. 184–199. [Online]. Available: http://mmlab.ie.cuhk.edu.hk/projects/SRCNN.html

[20] Z. Shao, J. Cai, P. Fu, L. Hu, and T. Liu, "Deep learning-based fusion of Landsat-8 and Sentinel-2 images for a harmonized surface reflectance product," *Remote Sens. Environ.*, vol. 235, Dec. 2019, Art. no. 111425, doi: 10.1016/j.rse.2019.111425.

[21] H. Fu, G. Sun, A. Zhang, B. Shao, J. Ren, and X. Jia, "Tensor singular spectrum analysis for 3-D feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403914, doi: 10.1109/TGRS.2023.3272669.

[22] Q. Zhao, L. Ji, Y. Su, Y. Zhao, and J. Shi, "SRSF-GAN: A super-resolution-based spatial fusion with GAN for satellite images with different spatial and temporal resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5408019, doi: 10.1109/TGRS.2023.3329115.

[23] Y. Song, H. Zhang, and L. Zhang, "Remote sensing image spatiotemporal fusion via a generative adversarial network through one prior image pair," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 7009–7012, doi: 10.1109/IGARSS39084.2020.9324101.

[24] P. Qin, H. Huang, H. Tang, J. Wang, and C. Liu, "MUSTFN: A spatiotemporal fusion method for multi-scale and multi-sensor remote sensing images based on a convolutional neural network," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 115, Dec. 2022, Art. no. 103113, doi: 10.1016/j.jag.2022.103113.

[25] W. Sun, J. Li, M. Jiang, and Q. Yuan, "Supervised and self-supervised learning-based cascade spatiotemporal fusion framework and its application," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 19–36, Sep. 2023, doi: 10.1016/j.isprsjprs.2023.07.022.

[26] H. Jiang, Y. Qian, G. Yang, and H. Liu, "MLKNet: Multi-stage for remote sensing image spatiotemporal fusion network based on a large kernel attention," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1257–1268, 2024, doi: 10.1109/JSTARS.2023.3338978.

[27] X. Hu et al., "HDNet: High-resolution dual-domain learning for spectral compressive imaging," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17521–17530, doi: 10.1109/CVPR52688.2022.01702.

[28] Y. Cai et al., "Mask-guided spectral-wise transformer for efficient hyperspectral image reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17481–17490, doi: 10.1109/CVPR52688.2022.01698.

[29] L. Wang et al., "Exploring sparsity in image super-resolution for efficient inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4915–4924, doi: 10.1109/CVPR46437.2021.00488.

[30] R. Liu, P. Wang, B. Du, and B. Qu, "Endmember bundle extraction based on improved multiobjective particle swarm optimization," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: 10.1109/LGRS.2023.3287919.

[31] S. Wang and F. Fan, "Thangka hyperspectral image super-resolution based on a spatial–spectral integration network," *Remote Sens.*, vol. 15, no. 14, p. 3603, Jul. 2023, doi: 10.3390/rs15143603.

[32] Z. Zhao et al., "CDDFuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion," 2022, *arXiv:2211.14461*.

[33] H. Su, V. Jampani, D. Sun, O. Gallo, E. Learned-Miller, and J. Kautz, "Pixel-adaptive convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11158–11167, doi: 10.1109/CVPR.2019.01142.

[34] Z. Wu, Z. Liu, J. Lin, Y. Lin, and S. Han, "Lite transformer with long-short range attention," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Apr. 2020, pp. 1–13.

[35] H. Liu, M. Shao, Y. Qiao, Y. Wan, and D. Meng, "Unpaired image super-resolution using a lightweight invertible neural network," *Pattern Recognit.*, vol. 144, Dec. 2023, Art. no. 109822, doi: 10.1016/j.patcog.2023.109822.

[36] G. Sun et al., "Large kernel spectral and spatial attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519915, doi: 10.1109/TGRS.2023.3292065.

[37] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Mar. 2002, doi: 10.1109/97.995823.

[38] R. Steele, "Peak signal-to-noise ratio formulas for multistage delta modulation with RC-shaped Gaussian input signals," *Bell Syst. Tech. J.*, vol. 61, no. 3, pp. 347–362, Mar. 1982, doi: 10.1002/j.1538-7305.1982.tb03410.x.

[39] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.



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